

JUNWHA HONG

Research Officer at the Agency for Defense Development, South Korea

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RESEARCH INTERESTS

Computer Architecture, Systems for AI, Reconfigurable Computing, System and Hardware Security.

EDUCATION

Ulsan National Institute of Science and Technology (UNIST)

Mar 2020 – Feb 2024

B.S. in Computer Science and Engineering, *summa cum laude*

GPA: 4.13/4.3 (98.3/100), Ranked 4th out of 296 students

PUBLICATIONS

- [1] **How Far Can PIMs Replace GPUs in Energy-Efficient Multi-Turn Agent Serving?**
Junwha Hong, Guseul Heo, Hyunsu Ye, Jongse Park, Olivia Hsu
IEEE International Symposium on Workload Characterization (IISWC), 2026; short paper.
- [2] **Understanding and Exploiting Diagonal Attention Sparsity in Autoregressive Image Generation**
Daeun Kim, Junwha Hong, Changhun Oh, Yoonsung Kim, Yoonhyeong Lee, Jongse Park
IEEE International Symposium on Workload Characterization (IISWC), 2026.
- [3] **Tortoise and Hare Guidance: Accelerating Diffusion Model Inference with Multirate Integration**
Yunghee Lee, Byeonghyun Pak, Junwha Hong, Hoseong Kim
Conference on Neural Information Processing Systems (NeurIPS), 2025.
- [4] **CMASan: Custom Memory Allocator-aware Address Sanitizer**
Junwha Hong, Wonil Jang, Mijung Kim, Lei Yu, Yonghwi Kwon, Yuseok Jeon
IEEE Symposium on Security and Privacy (S&P), 2025.

WORK EXPERIENCE

Agency for Defense Development (ADD)

Mar 2024 – Present

Research Officer for National Defense (First Lieutenant)

Manager: Dr. Eunjin Ko.

Research Collaboration (CMU & KAIST)

May 2025 – Present

Advisors: Prof. Olivia Hsu, Prof. Jongse Park

- Characterized the architectural tradeoffs of GPU–PIM systems for energy-efficient multi-turn LLM agent serving.
- Evaluated session-KV residency in PIM as a second-tier prefix cache to mitigate recomputation from GPU-side eviction.

Advisor: Prof. Yuseok Jeon.

- Designed and implemented the compiler and runtime infrastructure of CMASan, integrating Clang AST analysis, LLVM instrumentation, and allocator-aware runtime support.
- Analyzed custom memory allocators across 100+ C/C++ projects and developed a CodeQL-assisted screening workflow based on recurring allocator implementation patterns.
- Discovered 19 previously unknown memory bugs, including 7 CVEs: CVE-2023-7104, CVE-2023-7152, CVE-2023-7158, CVE-2024-8946, CVE-2024-8947, CVE-2024-8948, and CVE-2024-11235.
- Contributed to Google OSS-Fuzz, PHP, SQLite, MicroPython, libSQL, and AFL++.

Advisor: Prof. Yonghwi Kwon.

- Adapted CMASan to the OSS-Fuzz testing environment and fuzz-tested NGINX, Apache, gRPC, MySQL, and other applications.

HONORS

Top Prize (Minister of Science and ICT Award)	2022
Millitech Challenge, Republic of Korea.	

National Science & Technology Scholarship	2022
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Excellence Prize	2021
UNIST-POSTECH-KAIST Data Science Competition.	

SELECTED COURSEWORK

Tensor Processor Design for Image Recognition, Building Customized Computers, Parallel Computing, Computer Architecture, Cloud Computing.

TECHNICAL SKILLS & EXPERIENCE

Programming Languages: C/C++, Python, Scala, Java

Frameworks & Tools: LLVM, NVIDIA Nsight, vLLM/SGLang, Chisel, Vivado HLS, Docker, GDB scripting, AFLPlusPlus, CodeQL

Server Hardware: NVIDIA DGX A100, PCIe H100, DGX H200, DGX B300

Edge Hardware: NVIDIA Jetson, commercial edge NPUs, PYNQ-Z2, Arty A7-35T